ASSIGNMENT 16

Name: Rajdeep Das

Roll No: CSB25203

Subject: Advanced Programming

Program:

```c
#include <stdio.h>

#include <stdlib.h>

#include <pthread.h>

#include <unistd.h>


#define BUFFER_SIZE 5

#define ITEMS 10


int buffer[BUFFER_SIZE];

int count = 0;


pthread_mutex_t mutex;

pthread_cond_t not_full;

pthread_cond_t not_empty;


/* PRODUCER THREAD */

void *producer(void *arg)

{

   for (int i = 1; i <= ITEMS; i++)

   {

      pthread_mutex_lock(&mutex);


      /* wait if buffer is full */
```

```
        while (count == BUFFER_SIZE)
        {
            printf("Producer waiting, buffer full...\n");
            pthread_cond_wait(&not_full, &mutex);
        }

        buffer[count] = i;
        count++;

        printf("Producer produced item %d | Buffer count = %d\n", i, count);

        /* signal consumer that data is available */
        pthread_cond_signal(&not_empty);

        pthread_mutex_unlock(&mutex);

        sleep(1);
    }

    return NULL;
}

/* CONSUMER THREAD */
void *consumer(void *arg)
{
    for (int i = 1; i <= ITEMS; i++)
    {
        pthread_mutex_lock(&mutex);

        /* wait if buffer is empty */
        while (count == 0)
```

```
        {
            printf("Consumer waiting, buffer empty...\n");
            pthread_cond_wait(&not_empty, &mutex);
        }

        int item = buffer[count - 1];
        count--;

        printf("Consumer consumed item %d | Buffer count = %d\n", item, count);

        /* signal producer that space is available */
        pthread_cond_signal(&not_full);

        pthread_mutex_unlock(&mutex);

        sleep(2);
    }

    return NULL;
}

int main()
{
    pthread_t p, c;

    pthread_mutex_init(&mutex, NULL);
    pthread_cond_init(&not_full, NULL);
    pthread_cond_init(&not_empty, NULL);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);
```

```
    pthread_join(p, NULL);
    pthread_join(c, NULL);

    pthread_mutex_destroy(&mutex);
    pthread_cond_destroy(&not_full);
    pthread_cond_destroy(&not_empty);

    return 0;
}
```